

Farm Animal Demonstration

Java Lab Observation

1. Aim

To develop a Java program using a FarmAnimal class to store and display the name, place, food, and sound of farm animals using a constructor and display method.

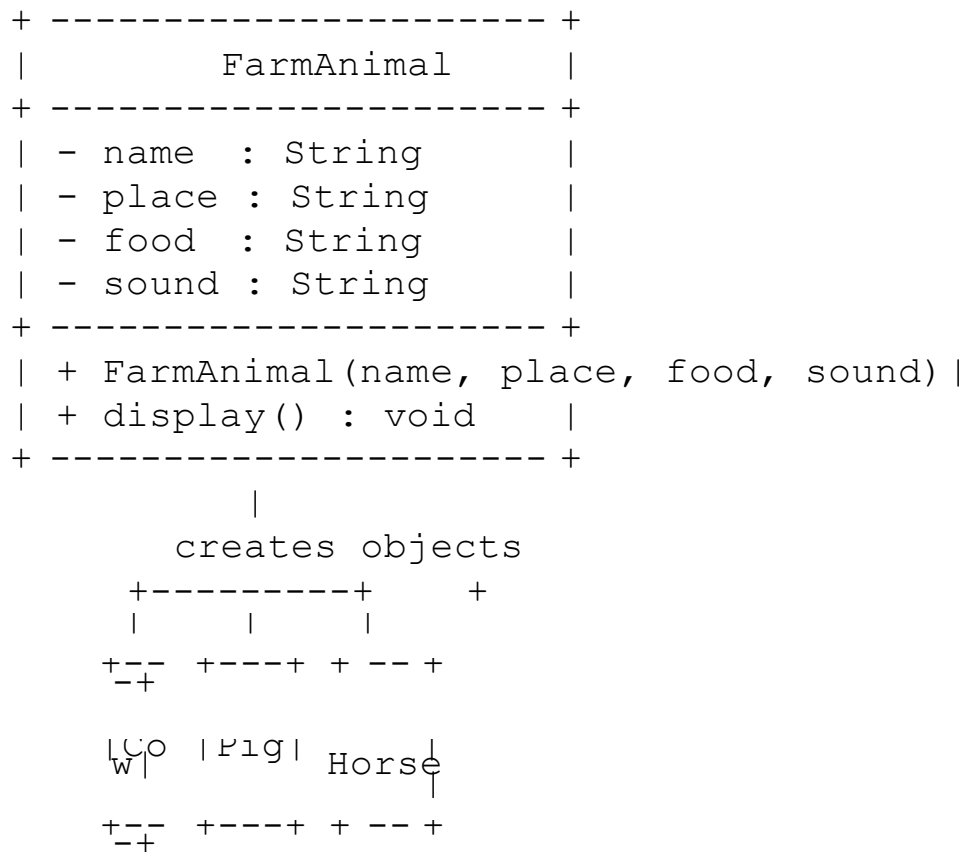
2. Step-by-Step Procedure

1. Create the FarmAnimal class.
2. Declare name, place, food, and sound as String variables.
3. Create a parameterized constructor to initialize all attributes.
4. Create the display() method to print animal details.
5. Create the FarmAnimalDemo class with main().
6. Create objects for Cow, Pig, and Horse with their details.
7. Call display() for each object.
8. Observe the output and end the program.

3. Algorithm

1. Start.
2. Define FarmAnimal with four String attributes.
3. Define a constructor to initialize the attributes.
4. Define display() to print the animal details.
5. Create Cow, Pig, and Horse objects.
6. Call display() for Cow, Pig, and Horse.
7. Stop.

4. UML Class Diagram



5. Code

```
class FarmAnimal {

    String
    name;
    String
    place;
    String
    food;
    String
    sound;

    // Constructor to initialize animal details
    FarmAnimal(String name, String place,
        String food, String sound) { this.name =
        name;
        this.place =
        place;
        this.food =
        food;
        this.sound =
        sound;
    }

    // Method to display
    animal details void
    display() {
        System.out.println("Animal
        Name : " + name);
        System.out.println("Stays
        At           : " +
        place);
        System.out.println("Eats :
        " + food);
        System.out.println("Sound
                        : " +
        sound);
        System.out.println();
    }
}

public class FarmAnimalDemo {
    public static void
    main(String[] args) {
        FarmAnimal cow = new
        FarmAnimal(
            "Cow", "Cowshed", "Grass", "Moo"
        );

        FarmAnimal pig = new FarmAnimal(
            "Pig", "Pigsty", "Vegetables", "Oink"
        );

        FarmAnimal horse = new
```

```
        FarmAnimal( "Horse",  
                    "Stable", "Grass",  
                    "Neigh"  
        );  
  
        cow.display();  
        pig.display();  
        horse.display()  
        ;  
    }  
}
```

6. Sample Output

```
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac FarmAnimalDemo.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java FarmAnimalDemo
Animal Name : Cow
Stays At    : Cowshed
Eats        : Grass
Sound       : Moo

Animal Name : Pig
Stays At    : Pigsty
Eats        : Vegetables
Sound       : Oink

Animal Name : Horse
Stays At    : Stable
Eats        : Grass
Sound       : Neigh
```

Output represented by the given code:

```
Animal Name : Cow
Stays At    : Cowshed
Eats        : Grass
Sound       : Moo

Animal Name : Pig
Stays At    : Pigsty
Eats        : Vegetables
Sound       : Oink

Animal Name : Horse
Stays At    : Stable
Eats        : Grass
Sound       : Neigh
```